
ME 521: Vehicle Design and Dynamics

Course Code	ME 521
Course Name	Vehicle Design and Dynamics
L-T-P-C	3-0-0-3
Prerequisites	Basic Engineering Mathematics and Mechanics
Intended for	M.Tech, MS, PhD, B.Tech 3rd/4th year students or consent of faculty
Mutual Exclusion	None
Approval	44th BoA (Nov 2021)

Course Contents

- Basic Vehicle Mechanics: Kinematics, Dynamics - Equation of motion, acceleration, effect of grade, drag. Air flow around the vehicle, tire models, rolling resistance. Load and energy calculations in electric vehicle over drive cycle. (6 hours)
- Vehicle Design: Roll cage design, camber and caster angle, Stability analysis. (6 hours)
- Transmission: Manual gear box, gear ratio, automatic gear box. Torque speed characteristics. Torque convertors, clutch, brakes. (8 hours)
- Road handling: Differentials system, Torque vectoring, Suspension. Double wishbone suspension system. (8 hours)
- Steering: Rack and Pinion steering system, Ackermann Steering, power steering. (6 hours)
- Vehicle Dynamics: Lumped mass modelling, Basics of vibration, Quarter car model, Longitudinal and Lateral dynamics. (10 hours)

Text Books

1. Gillespie, Thomas D., **Fundamentals of vehicle dynamics**, Vol. 400, Society of automotive engineers, 1992.
2. Rill, Georg, and Abel Arrieta Castro, **Road Vehicle Dynamics: Fundamentals and Modeling with MATLAB**, CRC Press, 2020.

References

1. Husain, Iqbal., **Electric and hybrid vehicles: design fundamentals**, CRC press, 2021.
2. Heisler, Heinz., **Advanced vehicle technology**, Elsevier, 2002.